

Model Distillation

Research Seminar

Danila Biktimirov

CUB

2026-02-26

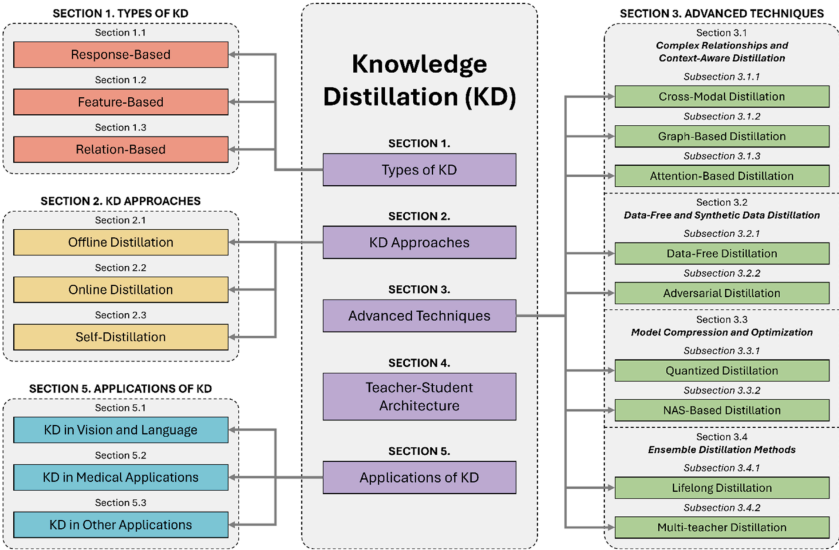
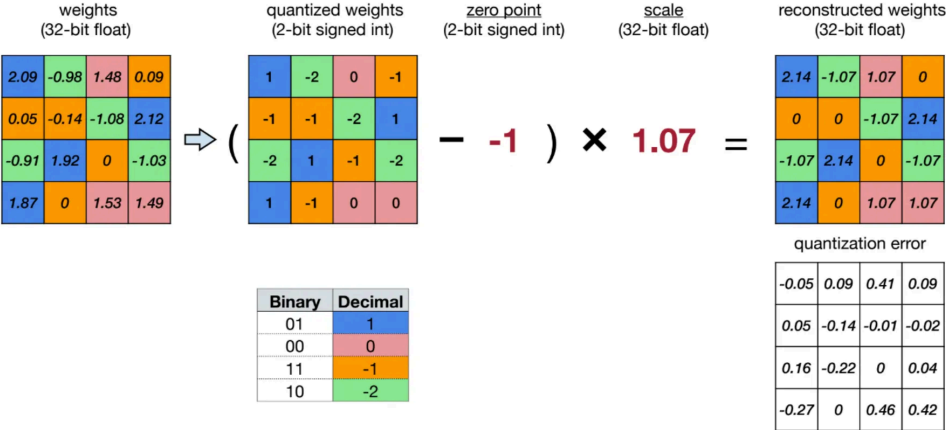


Fig. 2. The schematic structure of the paper and the relationship between its sections. The body of this survey mainly contains the fundamentals of knowledge distillation approaches, advanced techniques, and applications of knowledge distillation. Subsections of each main section are listed in this figure.



Motivation

SOTA models are too large and slow

Ensembles improve quality but complicate deployment

We want a solution for industrial use

The solution itself: distillation

Classical Distillation Method

Notation:

- N — number of samples
- K — number of classes
- L_{KD} — distillation loss function
- p_{ij} — class probabilities predicted by the student
- y_{ij} — class labels (hard labels)
- q_{ij} — class probabilities predicted by the teacher (soft labels)

$$\begin{aligned} L_{\text{KD}} &= \frac{1}{N} \sum_{i=1}^N \left(- \sum_{j=1}^K y_{ij} \log p_{ij} + \lambda D_{\text{KL}}(p_i \parallel q_i) \right) \\ &= \frac{1}{N} \sum_{i=1}^N \left(- \sum_{j=1}^K y_{ij} \log p_{ij} + \lambda \sum_{j=1}^K q_{ij} \log \left(\frac{q_{ij}}{p_{ij}} \right) \right) \\ &\approx - \frac{1}{N} \sum_{i=1}^N \left(\left(\sum_{j=1}^K y_{ij} \log p_{ij} + \lambda \sum_{j=1}^K q_{ij} \log p_{ij} \right) \right) \end{aligned}$$

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

Classical Distillation Method

$$q_i = \frac{\exp(\frac{z_i}{T})}{\sum_{j=1}^K \exp(\frac{z_j}{T})}$$

$$\frac{\partial C}{\partial z_i} = \frac{1}{T} (q_i - p_i) = \frac{1}{T} \left(\frac{e^{z_i/T}}{\sum_j e^{z_j/T}} - \frac{e^{v_i/T}}{\sum_j e^{v_j/T}} \right) \quad (2)$$

If the temperature is high compared with the magnitude of the logits, we can approximate:

$$\frac{\partial C}{\partial z_i} \approx \frac{1}{T} \left(\frac{1 + z_i/T}{N + \sum_j z_j/T} - \frac{1 + v_i/T}{N + \sum_j v_j/T} \right) \quad (3)$$

If we now assume that the logits have been zero-meaned separately for each transfer case so that $\sum_j z_j = \sum_j v_j = 0$ Eq. 3 simplifies to:

$$\frac{\partial C}{\partial z_i} \approx \frac{1}{NT^2} (z_i - v_i) \quad (4)$$

$$L_{\text{KD}} = -\frac{1}{N} \sum_{i=1}^N \left(\sum_{j=1}^K y_{ij} \log p_{ij} + \lambda T^2 \sum_{j=1}^K q_{ij}^T \log p_{ij}^T \right)$$

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

Experiments with MNIST

Experiment 1:

- Teacher: 67 errors
- Student without distillation: 144 errors
- Student with distillation: 74 errors

Experiment 2:

- Without one class (3)
- 206 test errors, among which 133 are threes out of 1010 threes in MNIST
- If to add 3.5 to the bias, the model makes 109 errors, among which only 14 are threes and accuracy is 98.6%

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

Distillation in Speech Recognition

- Mapping an audio signal to text
- 8 layers
- 2560 neurons
- 85 million parameters
- 10 models in the ensemble

System	Test Frame Accuracy	WER
Baseline	58.9%	10.9%
10× Ensemble	61.1%	10.7%
Distilled Single Model	60.8%	10.7%

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

Specialist Ensembles

JFT Dataset

- Internal Google dataset
- 100 million images
- 15,000 classes
- The base model was trained for 6 months
- Training an ensemble of models would take years

Solution

- Introduce one shared model
- Many specialists for similar classes

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

Specialist Ensembles

System	Conditional Test Accuracy	Test Accuracy
Baseline	43.1%	25.0%
+61 Specialist models	45.9%	26.1%

# of specialists covering	# of test examples	delta in top1 correct	relative accuracy change
0	350037	0	0.0%
1	141993	+1421	+3.4%
2	67161	+1572	+7.4%
3	38801	+1124	+8.8%
4	26298	+835	+10.5%
5	16474	+561	+11.1%
6	10682	+362	+11.3%
7	7376	+232	+12.8%
8	4703	+182	+13.6%
9	4706	+208	+16.6%
10 or more	9082	+324	+14.1%

Table 4: Top 1 accuracy improvement by # of specialist models covering correct class on the JFT test set.

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

Soft Targets as Regularizers

System & training set	Train Frame Accuracy	Test Frame Accuracy
Baseline (100% of training set)	63.4%	58.9%
Baseline (3% of training set)	67.3%	44.5%
Soft Targets (3% of training set)	65.4%	57.0%

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network.

On the one hand:

- SOTA models are huge
- Accuracy is high
- Require expensive hardware

On the other hand:

- In practice, small models are used
- ResNet-50, MobileNet
- Accuracy is lower, but they are fast and cheap to operate

Fact: ResNet-50 is downloaded 10× more often than larger BiT models!

Goal of this paper

To analyze already existing methods

Two Approaches to Compression

1. Pruning:

- Removes parts of a trained model
- Problems:
 - Cannot change the model family (ResNet → MobileNet)
 - Architectural difficulties (GroupNorm, rebalancing)

2. Distillation:

- Transfer of knowledge from teacher to student
- The architecture can be changed!

Proposals of This Paper

1. Consistency

- Teacher and student see the same crop/augmentation
- No precomputation (no fixed teacher!)

2. Aggressiveness

- Aggressive mixup
- Creation of “synthetic” data outside the natural diversity

3. Patience

- Very long training (thousands of epochs)
- On ImageNet: 9600 epochs

Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

A good teacher is patient and consistent

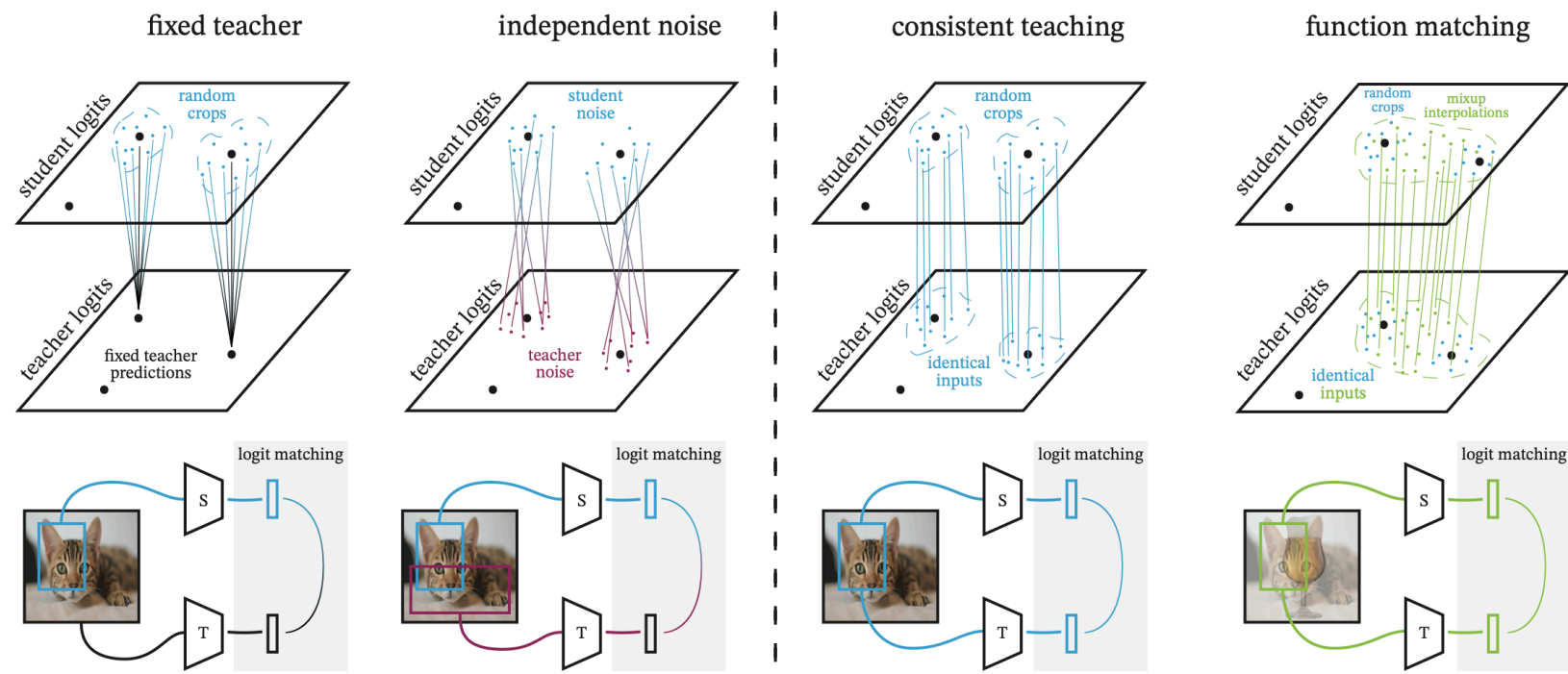


Figure 2. Schematic illustrations of various design choices when doing knowledge distillation. **Left:** *Teacher* receives a fixed image, while *student* receives a random augmentation. **Center-left:** *Teacher* and *student* receive independent image augmentations. **Center-right:** *Teacher* and *student* receive consistent image augmentations. **Right:** *Teacher* and *student* receive consistent image augmentations plus the input image manifold is extended by including linear segments between pairs of images (known as *mixup* [52] augmentation).

Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Experiments

Data:

- 5 datasets: Flowers102, Pets, Food101, SUN397, ImageNet
- From 1,020 to 1.28M images and from 37 to 1000 classes

Models:

- Teacher: BiT-ResNet-152x2 (pretrained on ImageNet-21k)
- Student: ResNet-50 (with GroupNorm instead of BatchNorm)

Loss Function:

- KL divergence (soft targets only)
- With temperature T (as in Hinton)

Training:

- Adam, cosine schedule, gradient clipping
- Mixup with aggressive coefficients

Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Importance of Consistency

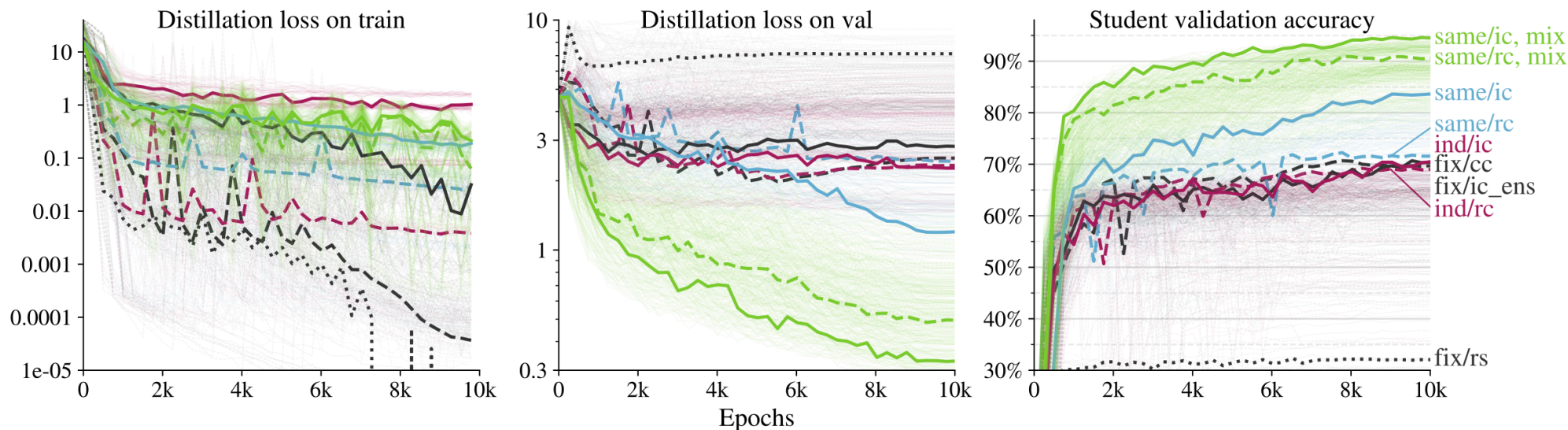


Figure 3. Experimental validation of the “consistency” requirement on the *Flowers102* dataset. Colors match different knowledge distillation design choices as introduced in Figure 2 and Section 3.1.1. Note that while the fixed teacher settings achieve significantly lower distillation loss, they lead to students which do not generalize well. In contrast, **consistent teaching** and **function matching** approaches lead to significantly higher student performance. Similar results on more datasets are reported in Appendix C.

Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Importance of Patience

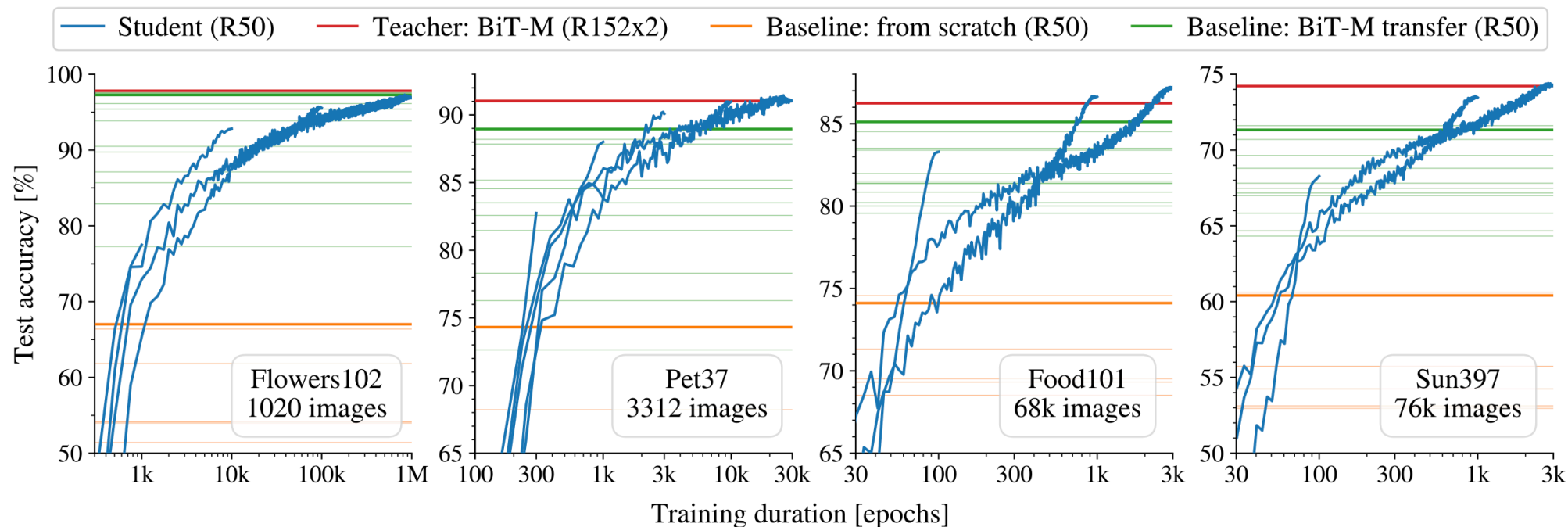


Figure 4. One needs patience along with consistency when doing distillation. Eventually, the teacher will be matched; this is true across various datasets of different scale.

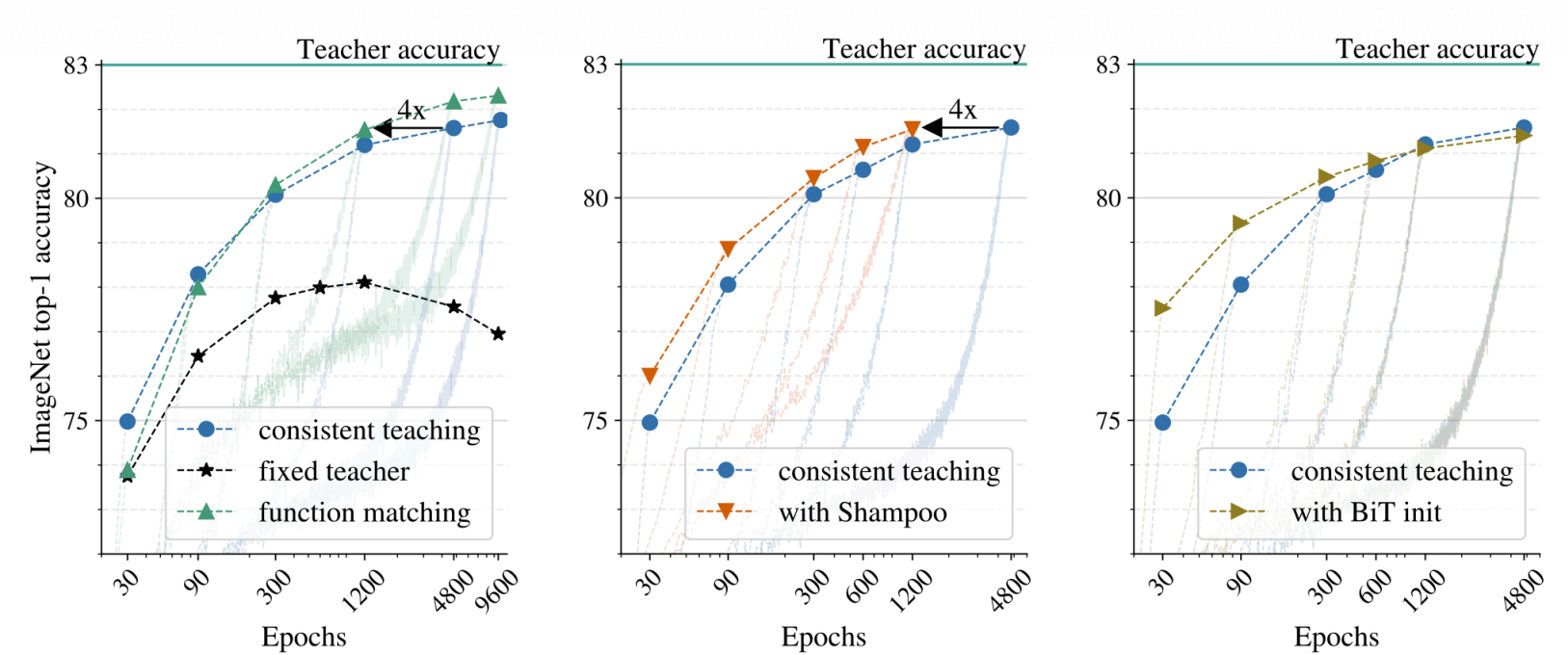
Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

SoTA Compression

Model	Arch.	Res.	Accuracy
“Revisiting ResNet” [2]	R50	160	78.8%
FunMatch (T224)	R50	160	80.5%
Original ResNet [12]	R50	224	77.2%
BiT-M-R50 [23]	R50	224	78.4%
Meal-v2 [38]	R50	224	80.7%
FunMatch (T384+224)	R50	224	82.8%
“Revisiting ResNet” [2]	R152	224	82.8%

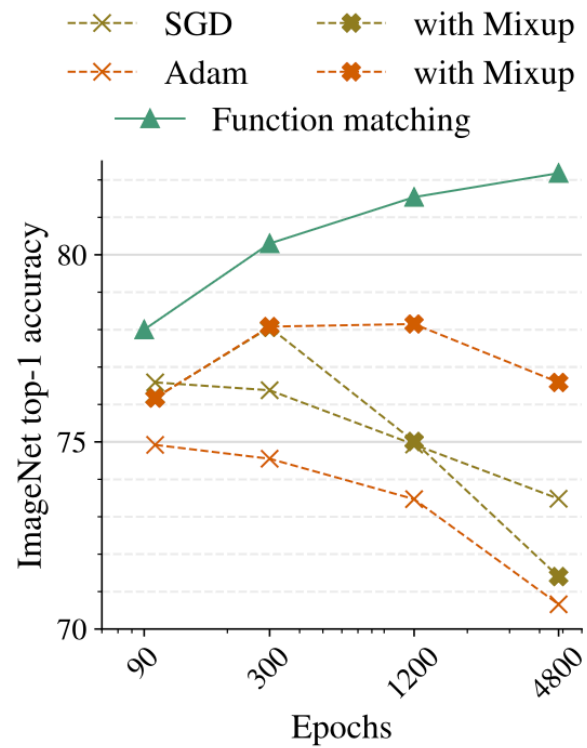
Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

ImageNet Results



Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Function Matching



Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Reasons of Bad Distillation

1. It is bad to precompute the teacher (for computational savings), because consistency is lost, leading to poor results.
2. Training is too short
 - Distillation requires 10–100× more epochs

Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Conclusions

1. The knowledge distillation process should be viewed as a function approximation problem.
2. It is important to ensure that the teacher and student models receive the same and sufficiently diverse input data.
3. To achieve better student model performance, a large number of epochs should be used — significantly more than in standard training.

Beyer, L., Zhai, X., Royer, A., Markeeva, L., Anil, R., & Kolesnikov, A. (2022). Knowledge distillation: A good teacher is patient and consistent.

Distillation in Language Models

Table 1: **DistilBERT retains 97% of BERT performance.** Comparison on the dev sets of the GLUE benchmark. ELMo results as reported by the authors. BERT and DistilBERT results are the medians of 5 runs with different seeds.

Model	Score	CoLA	MNLI	MRPC	QNLI	QQP	RTE	SST-2	STS-B	WNLI
ELMo	68.7	44.1	68.6	76.6	71.1	86.2	53.4	91.5	70.4	56.3
BERT-base	79.5	56.3	86.7	88.6	91.8	89.6	69.3	92.7	89.0	53.5
DistilBERT	77.0	51.3	82.2	87.5	89.2	88.5	59.9	91.3	86.9	56.3

Table 2: **DistilBERT yields to comparable performance on downstream tasks.** Comparison on downstream tasks: IMDB (test accuracy) and SQuAD 1.1 (EM/F1 on dev set). D: with a second step of distillation during fine-tuning.

Model	IMDb (acc.)	SQuAD (EM/F1)
BERT-base	93.46	81.2/88.5
DistilBERT	92.82	77.7/85.8
DistilBERT (D)	-	79.1/86.9

Table 3: **DistilBERT is significantly smaller while being constantly faster.** Inference time of a full pass of GLUE task STS-B (sentiment analysis) on CPU with a batch size of 1.

Model	# param. (Millions)	Inf. time (seconds)
ELMo	180	895
BERT-base	110	668
DistilBERT	66	410

Sanh, V., Debut, L., Chaumond, J., Wolf, T., & Rush, A. M. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter.

Distillation in Generative Language Models

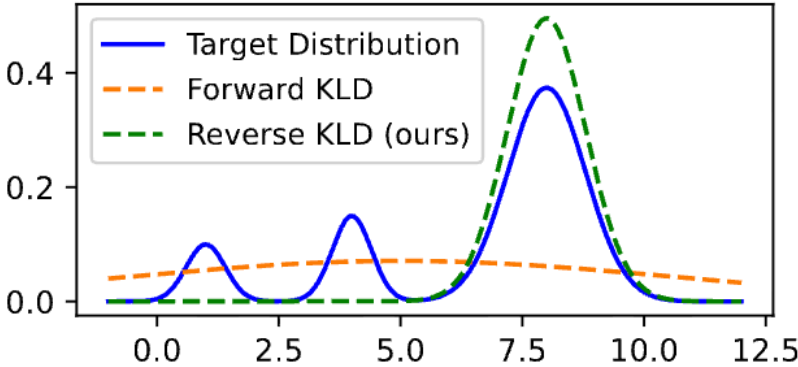
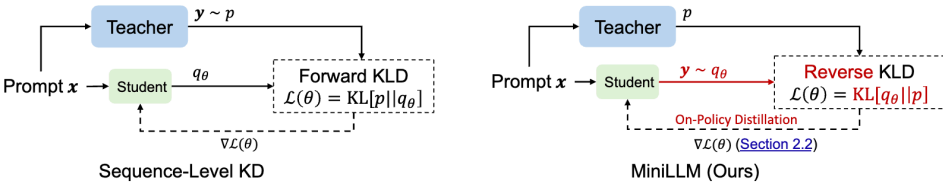


Figure 2: The toy experiment. We fit a Gaussian mixture distribution with a single Gaussian distribution using *forward* KLD and *reverse* KLD.



Gu, Y., Dong, L., Wei, F., Huang, M., et al. (2023). MiniLLM: On-Policy Distillation of Large Language Models

Distillation in Generative Language Models

$$L_{\text{OD}}(\theta) := E_{x \sim X} \left[E_{y \sim p_{S(\cdot | x)}} \left[D_{\text{KL}}(p_T \parallel p_S^\theta)(y | x) \right] \right]$$

Algorithm 1 Generalized Knowledge Distillation (GKD)

```
1: Given: Teacher model  $p_T$ , Student Model  $p_S^\theta$ , Dataset  $(X, Y)$  containing (input, output) pairs
2: Hyperparameters: Student data fraction  $\lambda \in [0, 1]$ , Divergence  $\mathcal{D}$ , Learning rate  $\eta$ 
3: for each step  $k = 1, \dots, K$  do
4:   Generate a random value  $u \sim \text{Uniform}(0, 1)$ 
5:   if  $u \leq \lambda$  then
6:     Sample inputs  $x$  from  $X$  and generate outputs  $y \sim p_S^\theta(\cdot | x)$  to obtain  $B = \{(x_b, y_b)\}_{b=1}^B$ 
7:   else
8:     Sample batch of inputs and outputs from  $(X, Y)$  to obtain  $B = \{(x_b, y_b)\}_{b=1}^B$ .
9:   end if
10:  Update  $\theta$  to minimize  $L_{\text{GKD}}$ :  $\theta \leftarrow \theta - \eta \frac{1}{B} \sum_{(x,y) \in B} \nabla_\theta \mathcal{D}(p_T \parallel p_S^\theta)(y|x)$ 
11: end for
```

$$L_{\text{GKD}}(\theta) := (1 - \lambda) E_{(x,y) \sim (X,Y)} \left[D(p_T \parallel p_S^\theta)(y | x) \right] + \lambda E_{x \sim X} \left[E_{y \sim p_{S(\cdot | x)}} \left[D(p_T \parallel p_S^\theta)(y | x) \right] \right]$$

Agarwal, R., Vieillard, N., Zhou, Y., Stanczyk, P., Ramos, S., Geist, M., Bachem, O. (2023). On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes

Gemini 1.5 Flash (1)

Gemini 1.5 Pro	Relative to 1.5 Pro (Feb)	Relative to 1.0 Pro	Relative to 1.0 Ultra
Long-Context Text, Video & Audio	no change	from 32k up to 10M tokens	from 32k up to 10M tokens
Core Capabilities	Win-rate: 78.1% (25/32 benchmarks)	Win-rate: 88.0% (44/50 benchmarks)	Win-rate: 77.8% (35/45 benchmarks)
Text	Win-rate: 78.6% (11/14 benchmarks)	Win-rate: 95.8% (23/24 benchmarks)	Win-rate: 84.2% (16/19 benchmarks)
Vision	Win-rate: 92.3% (12/13 benchmarks)	Win-rate: 95.2% (20/21 benchmarks)	Win-rate: 85.7% (18/21 benchmarks)
Audio*	Win-rate: 80% (4/5 benchmarks)	Win-rate: 60% (3/5 benchmarks)	Win-rate: 40% (2/5 benchmarks)

Google DeepMind. (2024). Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context.

Gemini 1.5 Flash (2)

Gemini 1.5 Flash	Relative to 1.0 Pro	Relative to 1.0 Ultra
Long-Context Text, Video & Audio	from 32k up to 10M tokens	from 32k up to 10M tokens
Core Capabilities	Win-rate: 82.0% (41/50 benchmarks)	Win-rate: 46.7% (21/44 benchmarks)
Text	Win-rate: 94.7% (18/19 benchmarks)	Win-rate: 42.1% (8/19 benchmarks)
Vision	Win-rate: 90.5% (19/21 benchmarks)	Win-rate: 61.9% (13/21 benchmarks)
Audio	Win-rate: 0% (0/5 benchmarks)	Win-rate: 0% (0/5 benchmarks)

Google DeepMind. (2024). Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context.

Gemini 1.5 Flash (3)

Language	Gemini 1.5 Flash	Gemini 1.5 Pro	GPT-3.5 Turbo	GPT-4 Turbo	Claude 3 Haiku	Claude 3 Sonnet	Claude 3 Opus
English	1.5	4.3	2.6	6.8	2.2	6.2	10.5
Japanese	4.3	10.9	12.9	35.4	10.7	23.9	46.6
Chinese	5.2	14.1	18.4	53.4	12.8	30	55.3
French	1.9	4.7	4.1	10.7	3.0	6.8	13.4

Table 3 | Time per output character (ms) of various APIs for English, Japanese, Chinese, and French responses, given inputs of 10,000 characters. Gemini 1.5 Flash achieves the fastest output generation for all languages tested.

Google DeepMind. (2024). Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context.

References

1. Hinton, G., Vinyals, O., & Dean, J. (2015). *Distilling the Knowledge in a Neural Network*.
2. Yandex Handbook — Knowledge Distillation
3. Beyer, L., Zhai, X., Royer, A., et al. (2022). *Knowledge Distillation: A Good Teacher is Patient and Consistent*.
4. Gu, Y., Dong, L., Wei, F., Huang, M., et al. (2023). *MiniLLM: On-Policy Distillation of Large Language Models*
5. Agarwal, R., Vieillard, N., Zhou, Y., Stanczyk, P., Ramos, S., Geist, M., Bachem, O. (2023). *On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes*
6. Google DeepMind. (2024). *Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context*.
7. Hsieh, C.-Y., et al. (2023). *Distilling Step-by-Step! Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes*.
8. Sanh, V., Debut, L., Chaumond, J., Wolf, T., & Rush, A. M. (2019). *DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter*.